

2016.0 RANGE ROVER (LG), 501-29

SIDE PANEL SHEET METAL REPAIRS

SIDE PANEL FRONT SECTION

[G1454616]

REMOVAL AND INSTALLATION

REMOVAL



All vehicles

NOTES:

- The side panel front section is installed in conjunction with:
 - Front fender
 - Front fender lower mounting bracket
 - Fender splash shield
 - Fender apron panel
 - Windshield
 - Instrument panel
 - Front door
 - Side air curtain module
- Right hand indicated, left hand similar.

1.



E143339

The side panel front section is serviced as indicated.

-
2. Before commencing this procedure make sure that you are aware of all Health and Safety requirements.

For additional information, refer to: [Health and Safety Precautions](#) (100-00 General Information, Description and Operation).

-
3. For further information on the methods, tools and fixings used in this procedure refer to the body repairs - general information section.

NOTES:

- Where breakstem fasteners (BSF) removal is instructed remove fastener shaft with a 4mm punch and remove the head with a 6.5mm drill bit. Make sure all the drilled holes are free from debris.
- Where self piercing rivets (SPR) removal is instructed use either the approved SPR tool or 5.3mm drill bit.
- Any joints sealed as part of vehicle manufacture must be sealed as part of the repair.

-
4. For additional information, refer to: [Fender](#) (501-02 Front End Body Panels, Removal and Installation).

 5. For additional information, refer to: [Front Fender Lower Mounting Bracket](#) (501-27 Front End Sheet Metal Repairs, Removal and Installation).

 6. For additional information, refer to: [Fender Splash Shield](#) (501-02 Front End Body Panels, Removal and Installation).

 7. For additional information, refer to: [Fender Apron Panel](#) (501-27 Front End Sheet Metal Repairs, Removal and Installation).

 8. For additional information, refer to: [Windshield Glass](#) (501-11 Glass, Frames and Mechanisms, Removal and Installation).

 9. For additional information, refer to: [Instrument Panel Lower Section](#) (501-12B Instrument Panel and Console - Long Wheel Base, Removal and Installation).

 10. For additional information, refer to: [Instrument Panel Upper Section](#) (501-12B Instrument Panel and Console - Long Wheel Base, Removal and Installation).

 11. For additional information, refer to: [Front Door](#) (501-03 Body Closures, Removal and Installation).

 12. For additional information, refer to: [Side Air Curtain Module](#) (501-20B Supplemental Restraint System, Removal and Installation).



Vehicles with roof opening panel

-
13. For additional information, refer to: [Roof Front Panel](#) (501-28 Roof Sheet Metal Repairs, Removal and Installation).



Vehicles without roof opening panel

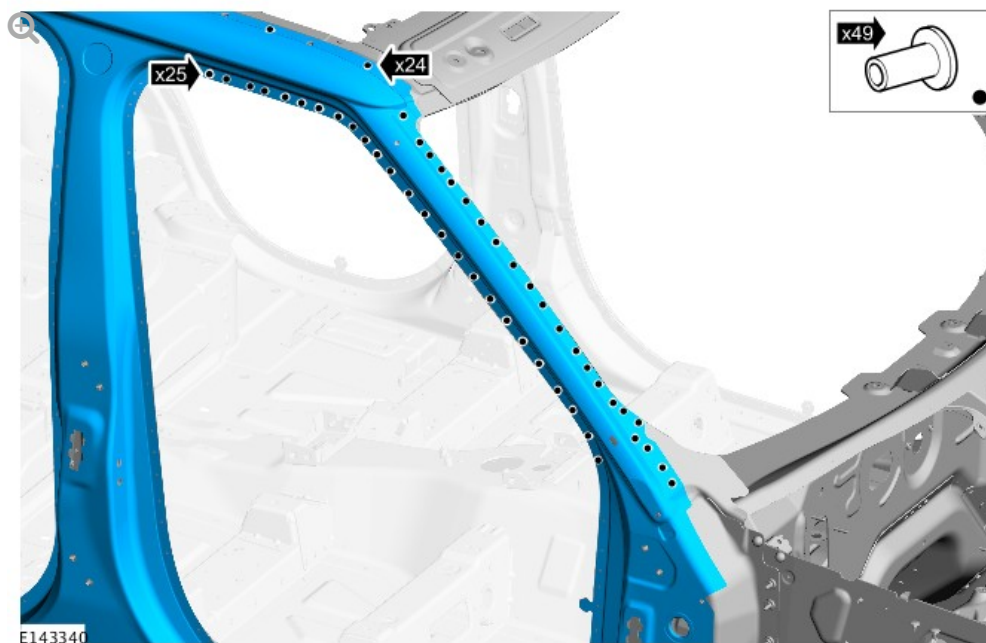
14. For additional information, refer to: [Roof Panel](#) (501-28 Roof Sheet Metal Repairs, Removal and Installation).



All vehicles

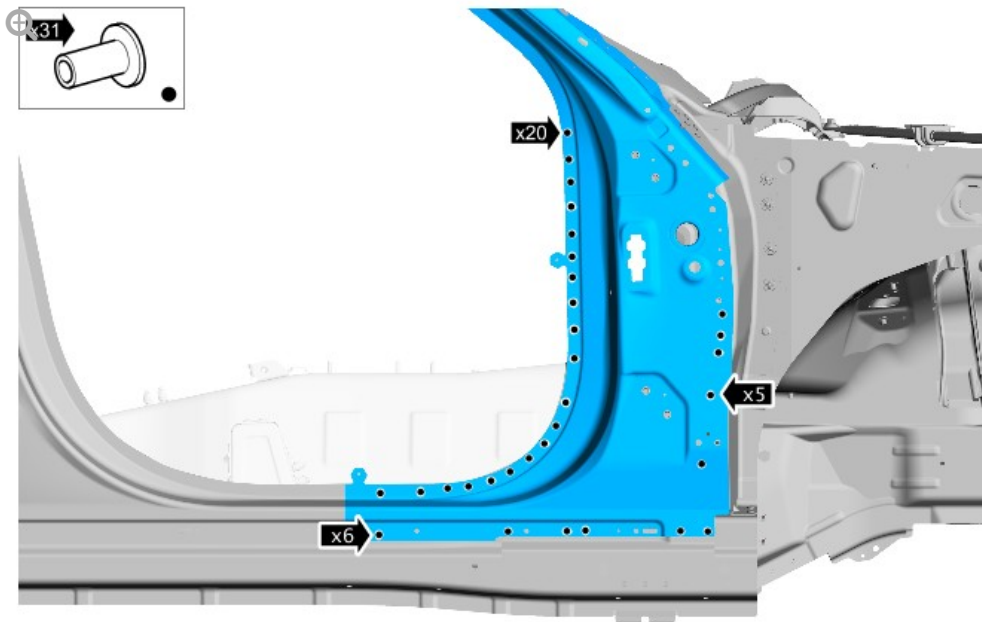
15. Release the side panel front section wiring harness and position it to one side.

16.



Remove the SPR's as indicated.

17.



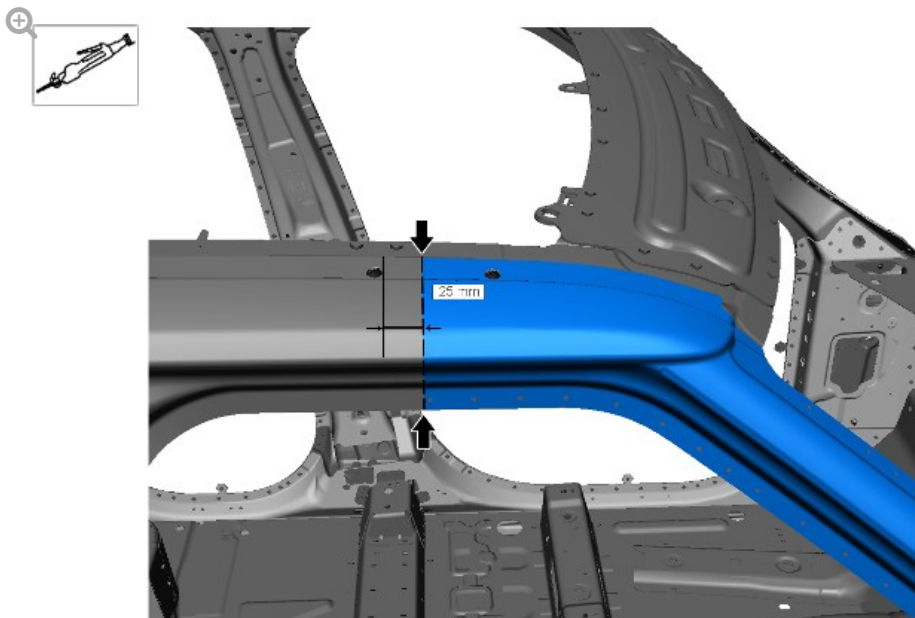
E143341

Remove the SPR's as indicated.

18.

NOTE:

Care should be taken not to cut through into the inner panel.



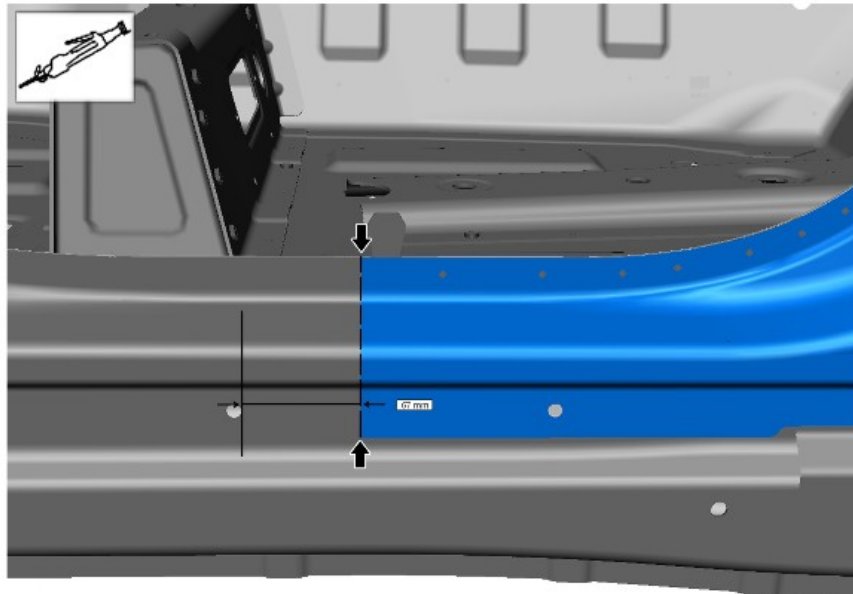
E143342

Mark out the position where the side panel front section MIG (metal inert gas) butt joint is to be made. Cut through the side panel front section as indicated.

19.

NOTE:

Care should be taken not to cut through into the inner panel.



E143343

Mark out the position where the side panel front section MIG butt joint is to be made. Cut through the side panel front section as indicated.

20. Separate the joints and remove the side panel front section.

INSTALLATION

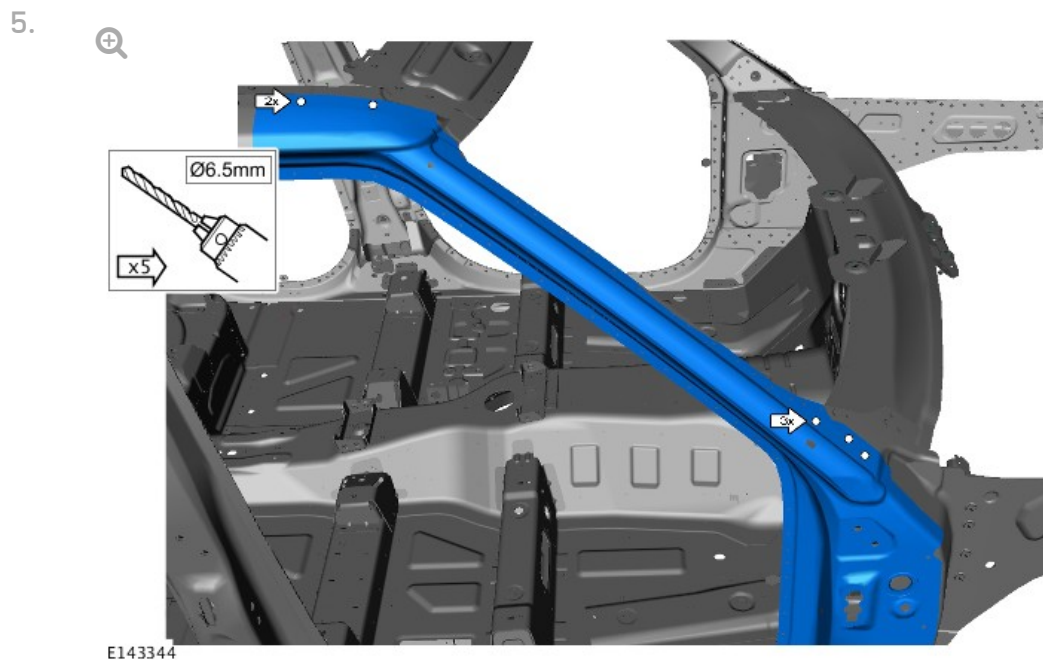
CAUTION:

Where structural adhesive is to be applied between surfaces it is essential compression is applied to the joints during the curing period.

NOTES:

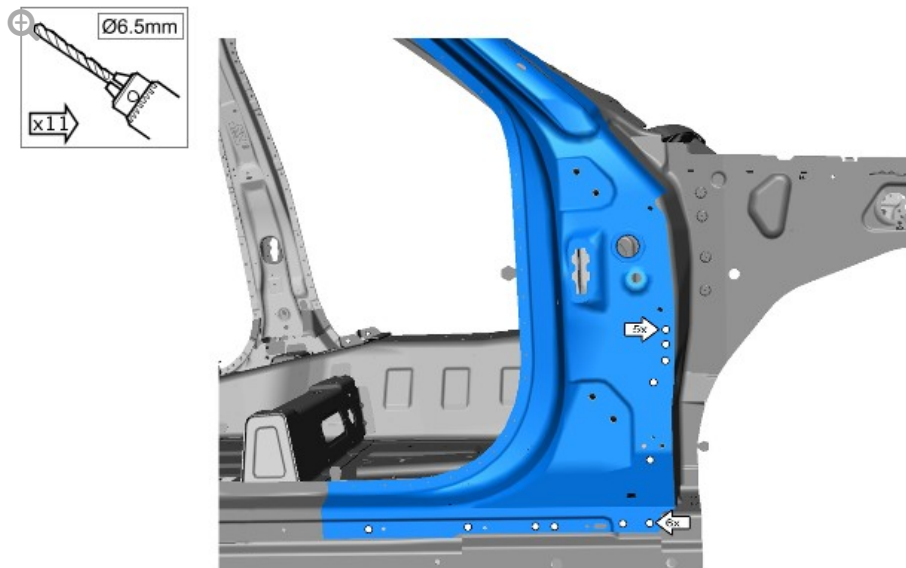
- BSF installation involves using the Land Rover approved BSF installation tool and requires a hole diameter of 6.5mm.
- SPR installation requires using the Land Rover approved SPR tool.

1. Remove rivet remnants.
2. Dress flanges where necessary.
3. Using a fine bristle disc, clean and prepare the panel surfaces.
4. Offer up the new side panel front section and clamp into position. Check alignment, if correct proceed to next step, if not, rectify and recheck before proceeding.



Drill holes through the old side panel front section panel into the new side panel front section where BSF's are to be installed as indicated.

6.

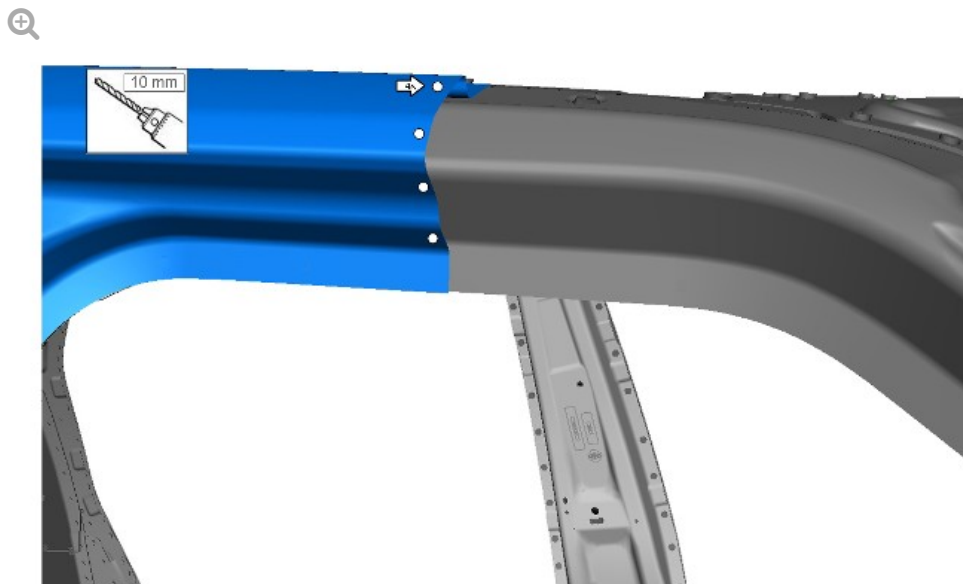


E143345

Drill holes through the old side panel front section into the new side panel front section where BSF's are to be installed as indicated.

7. Remove the new side panel front section.

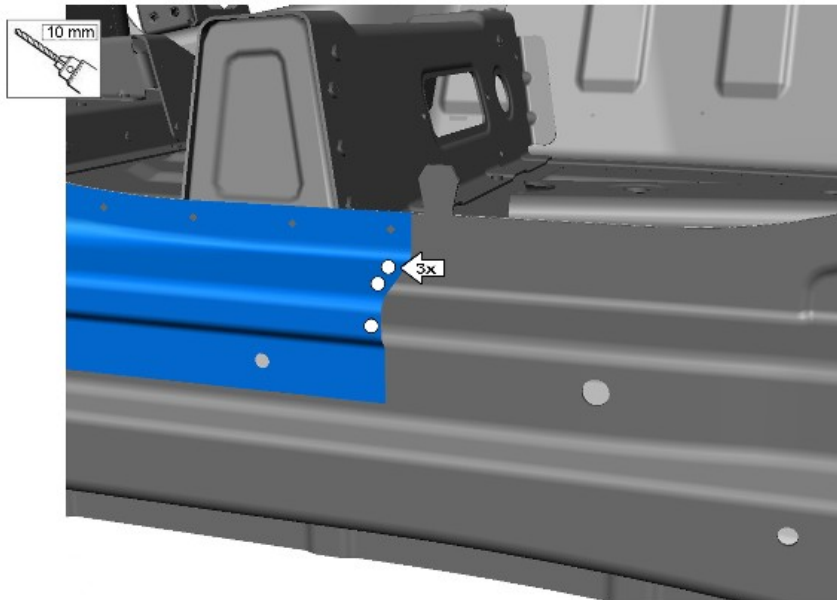
8.



E143346

Using a 10mm drill bit, drill holes for the MIG plug welds ready for the installation of the backing plate as indicated.

9.

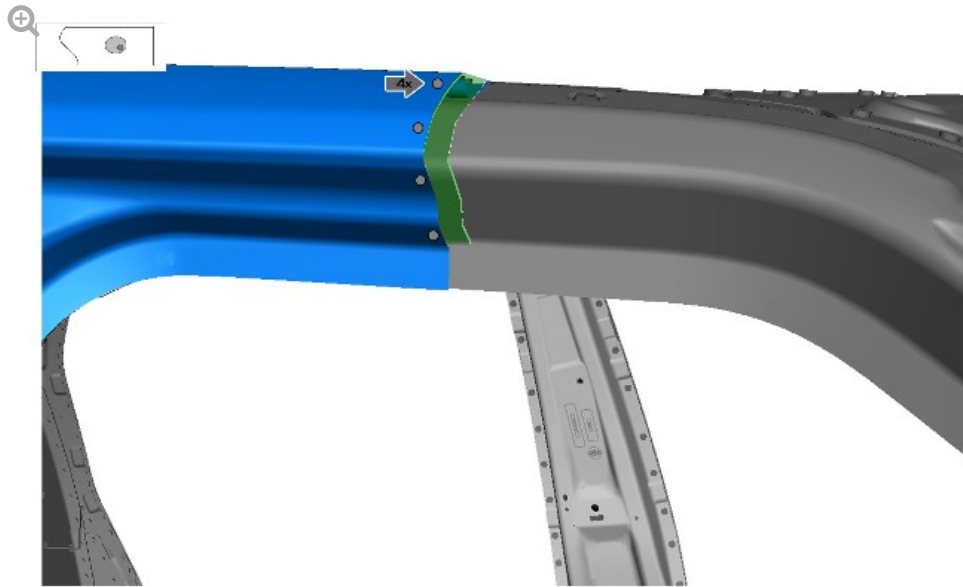


E143347

Using a 10mm drill bit, drill holes for the MIG plug welds ready for the installation of the backing plate as indicated.

-
10. Deburr the drilled holes.
-
11. Measure and cut two 50mm wide backing plate, (25mm each side of the MIG butt joint) from the side panel front section remnant.
-
12. Cut four run on/run off tab from the side panel front section remnant.
-
13. Using a fine bristle disc, clean and prepare the panel surfaces.
-

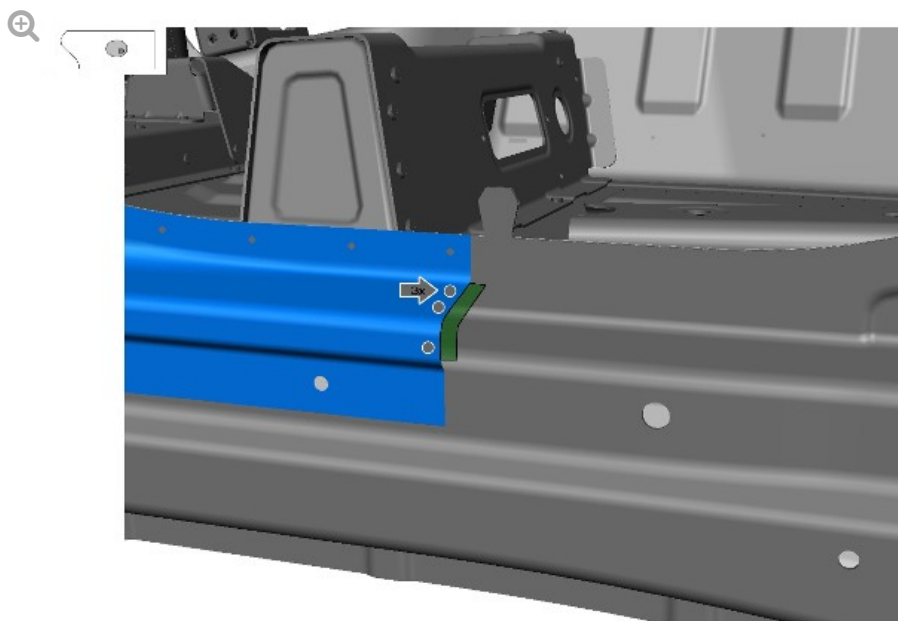
14.



E143348

Install and MIG plug weld the backing plate as indicated.

15.



E144298

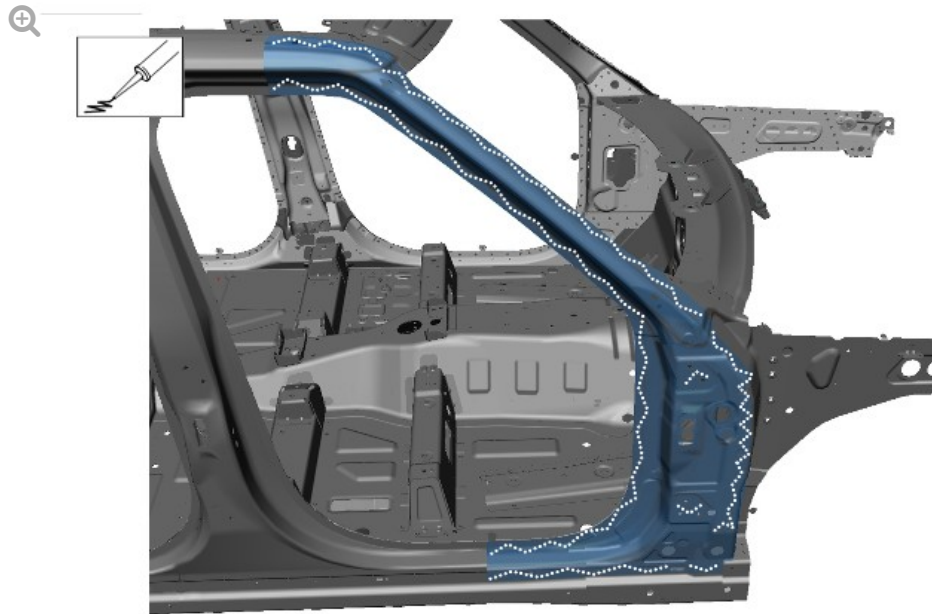
Install and MIG plug weld the backing plate as indicated.

-
16. Apply the coupling agent where the Land Rover recommended bonding material is to be applied and allow to dry.
-

17.

NOTE:

Make sure a continuous bead of adhesive surrounds fixing holes.

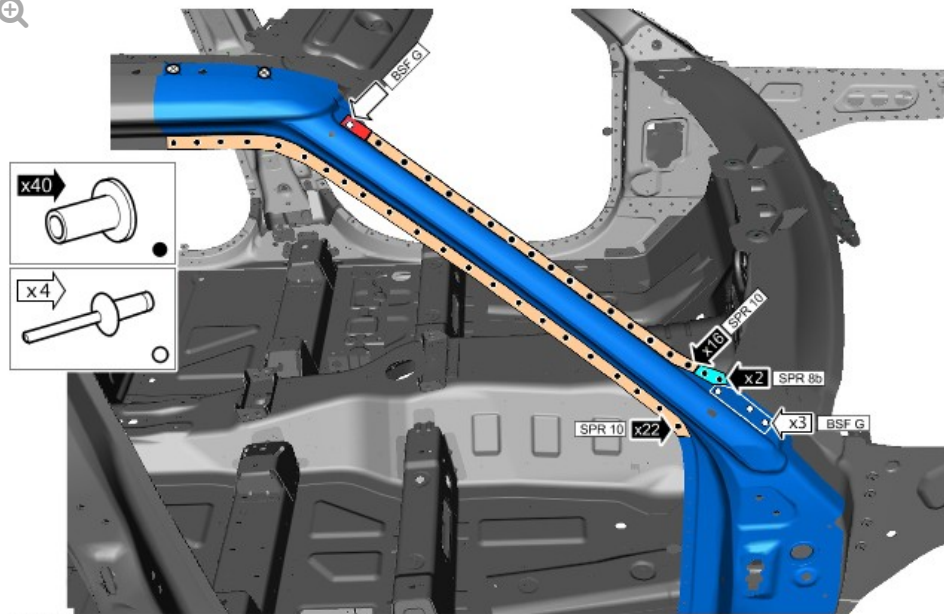


E144300

Apply a 5mm zig zag bead of the Land Rover recommended bonding material as indicated.

-
18. Offer up the new panel and clamp into position. Check alignment, if correct proceed to next step, if not, rectify and recheck before proceeding.
-

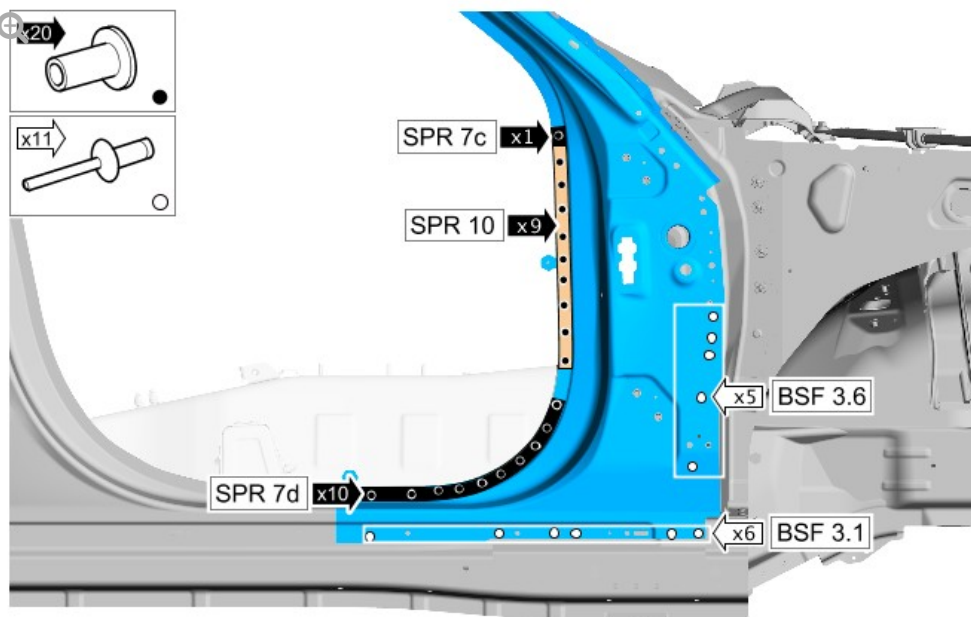
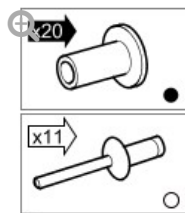
19.



E144301

Install the SPR's and the BSF's as indicated.

20.



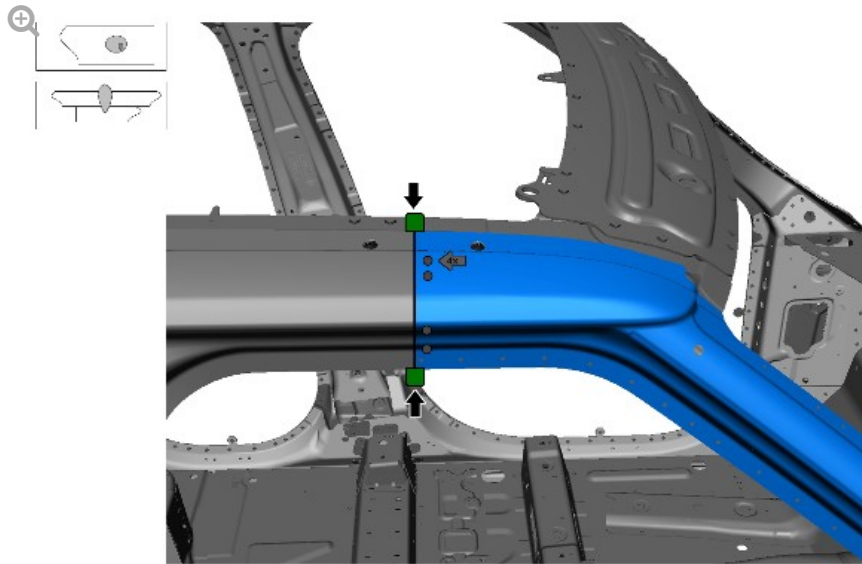
E144302

Install the SPR's and the BSF's as indicated.

21. Remove any excess adhesive.

22. Install and tack weld the run-on/run-off tabs to the side panel front section.

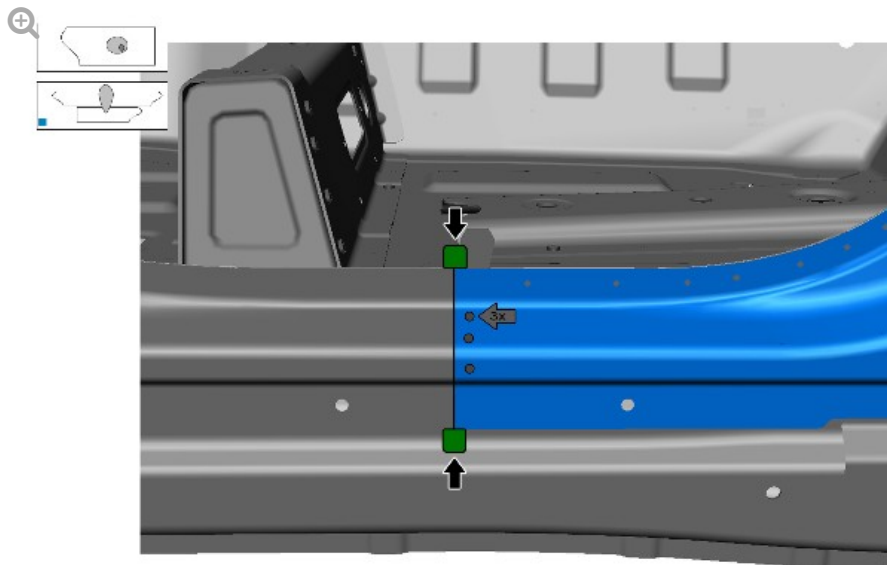
23.



E 144303

Install and MIG plug weld the backing plate and MIG weld the side panel front section butt joint as indicated.

24.



E 144304

Install and MIG plug weld the backing plate and MIG weld the side panel front section butt joint as indicated.

25. Cut off the run on/run off tabs.

26. Carry out a non destructive crack inspection on the MIG butt joints.

If correct proceed to next step, if not, rectify and recheck before proceeding.

27. Dress the welded joints.

28. Make sure that any open or exposed panel joints are suitably sealed.

29. Make sure corrosion protection is applied to all areas affected by repair.

30. The installation of associated panels and components is the reversal of removal procedure.